

# Separation of RNA from DNA with guanidinium thiocyanate

## Risk assessment

- Chloroform is a **KNOWN CARCINOGEN** and **REPRODUCTIVE TOXIN**!
  - Phenol is a **REPRODUCTIVE TOXIN** and causes serious chemical burns
  - Guanidine salts form **TOXIC GASES** with bleach or strong acid
- ▷ Work in a certified chemical fume hood
  - ▷ Wear gloves, safety glasses, lab coat
  - DO NOT add bleach or acidic solutions directly into sample waste



Reviewed: Jun 10, 2023

This is a bench card. Full protocol available online.

## >> Cell homogenization and RNA extraction

- |                                                 |                                                           |
|-------------------------------------------------|-----------------------------------------------------------|
| <input type="checkbox"/> R0136 Solution D, 2 mL | <input type="checkbox"/> R0141 Chloroform/Isoamyl alcohol |
| <input type="checkbox"/> R0140 Phenol, 1 mL     |                                                           |

- (1.) Add 1 mL Solution D per 100 mg fresh tissue (minced on ice and homogenized) or  $1 \times 10^7$  cultured cells. Homogenize by pipetting up and down at least ten times.

**Critical:** Do not thaw frozen tissue. Pulverize tissue samples under liquid nitrogen before adding Solution D. Remove medium from suspension or adherent cells before lysis. ←

- (2.) Incubate for 5 min at room temperature.

**Critical:** Solution D inactivates RNases. Do not store samples in Solution D longer than 30 min; freeze if pausing. ←

- (3.) Add 0.1 vol 2 M sodium acetate (pH 4.0). Mix thoroughly by inversion.

- (4.) Add 1.0 vol water-saturated phenol. Mix thoroughly by inversion.

**Critical:** Use acidic phenol. Buffered phenol will not separate RNA from DNA and proteins. ←

- (5.) Add 0.2 vol chloroform/isoamyl alcohol (49:1). Shake vigorously for 15 s by hand. Do not vortex.

**Critical:** Ensure caps are tightly closed during mixing. ←

- (6.) Cool samples on ice for 15 min.

- (7.) Centrifuge at  $10\,000 \times g$  for 20 min at 4 °C.

- (8.) Carefully transfer the colorless aqueous layer to a new tube. Avoid disturbing the interphase. If turbid, repeat chloroform extraction.

## >> RNA purification

- |                                            |                                              |
|--------------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Isopropyl alcohol | <input type="checkbox"/> R0172 Water, 100 mL |
| <input type="checkbox"/> 75% Ethanol       |                                              |

- (1.) Add an equal volume of isopropyl alcohol. Precipitate RNA on ice for 15–30 min.

- (2.) Centrifuge at  $12\,000 \times g$  for 15 min at 4 °C. Discard supernatant. Briefly spin again to remove residual solvent.

- (3.) Resuspend pellet in 300 µL Solution D.

- (4.) Add 300 µL isopropyl alcohol. Repeat precipitation on ice.

### *Separation of RNA from DNA with guanidinium thiocyanate*

- (5.) Centrifuge at  $12\,000 \times g$  for 10 min at 4 °C. Discard supernatant.
- (6.) Wash pellet with 600  $\mu\text{L}$  75% ethanol. Vortex briefly. Incubate 15 min at room temperature to remove guanidinium salts. ⌚ 15 min
- (7.) Centrifuge at  $12\,000 \times g$  for 10 min at 4 °C. Discard supernatant.
- (8.) Air-dry pellet for 10–15 min.

**Critical:** Do not overdry or vacuum-dry RNA—solubility will decrease. ←
- (9.) Dissolve RNA in 100–200  $\mu\text{L}$  DEPC-treated water or 0.5% SDS. Incubate at 60 °C for 10–15 min.
- (10.) Store RNA at –80 °C. ⚠

🔗 Recipe (available online) ⚠ Troubleshooting (available online) 📖 Notes (available online)

*Lab Protocols* — © B. C. Buchmuller. Licensed CC-BY-SA-4.0. Research use; provided as-is, without warranty. Current when printed. Visit <https://benjbuch.github.io/check/?q=a1fff60> or scan the QR code to read the full protocol or to check for updates.

